Huawei Access Node YEL-ACC-17 Hardware/Environmental Fault Resolution SOP

Node: YEL-ACC-17 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2200

Description

The failure of Node YEL-ACC-17, a Huawei Access Node located at the Yelahanka Sub-station, is attributed to a hardware/environmental fault within the Role: Access domain. Specifically, the occurrence of ASIC memory parity errors and line card crashes, coupled with the loss of power supply redundancy, precipitates the issuance of Alarm Code: ALARM_2200, classified as Critical. This Standard Operating Procedure (SOP) is designed to navigate the complex interplay of hardware and environmental factors influencing the Huawei Access Node's performance. It is essential to comprehend that the Huawei Access Node's architecture, which incorporates a combination of ASICs (Application-Specific Integrated Circuits) and line cards, is prone to memory parity errors due to the inherent limitations of silicon-based memory. Furthermore, the reliance on power supply redundancy underscores the vulnerability to environmental stressors such as power fluctuations or overheating. Consequently, a critical failure in this domain necessitates a multi-faceted approach, entailing diagnostic scrutiny and targeted resolution steps to restore the Node's operational integrity.

Pre-requisite Checks

- Pre-requisite: Verify the status of redundant links and confirm the presence of a standby Node for switchover, Description: Ensure that the redundant links are operational and ready for switchover to minimize service disruption.

- Pre-requisite: Ensure dispatch of on-site personnel for physical inspection and repair, Description: On-site personnel will assess the hardware and provide a detailed report of any damage or malfunction.

- Pre-requisite: Perform a thorough review of environmental conditions, including temperature and humidity levels, Description: Verify that the ambient temperature and humidity levels within the Sub-station are within the acceptable range for the Huawei Access Node.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: display device internal status, Expected Output: [Device ID] Internal Status: OK, Description: This command is used to verify the internal status of the device. A 'OK' status indicates that the device
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is functioning correctly.

Command: display board information, Expected Output: [Board ID] Board
Information: Normal, Description: This command is employed to verify
the status of the boards within the device. A 'Normal' status
signifies that the boards are functioning correctly.

Command: display power supply information, Expected Output: [Power
Supply ID] Power Supply Information: Normal, Description: This command
is utilized to verify the status of the power supply. A 'Normal'
status indicates that the power supply is functioning correctly.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC or line card, Command: replace {asic/line-card} {id}, Description: Replace the faulty ASIC or line card with a new one to restore the device's operational integrity.

Step 2: Step: Restore power supply redundancy, Command: power supply {id} enable, Description: Enable the power supply redundancy to ensure that the device can operate in the event of a power failure.

Step 3: Step: Perform system reboot, Command: system reboot, Description: Perform a system reboot to ensure that the changes take effect and the device is functioning correctly.

Post-Check Verification

```
Command: display device internal status, Expected Output: [Device ID]
Internal Status: OK, Description: Verify that the internal status of
the device is 'OK' to ensure that the resolution steps were
successful.

Command: display board information, Expected Output: [Board ID] Board
Information: Normal, Description: Verify that the status of the boards
within the device is 'Normal' to ensure that the resolution steps were
successful.

Command: display power supply information, Expected Output: [Power
Supply ID] Power Supply Information: Normal, Description: Verify that
the status of the power supply is 'Normal' to ensure that the
resolution steps were successful.
```

Roll-back Procedures

- Step: Restore original configuration, Command: system restore, Description: Restore the original configuration to ensure that the device returns to its previous state.

- Step: Revert changes, Command: revert changes, Description: Revert the changes made during the resolution process to ensure that the device returns to its previous state.

- Step: Notify NOC, Command: , Description: Notify the Network Operations Center (NOC) of the failed resolution attempt and the actions taken to revert the changes.